

Exercise 2.14. Let a and b be positive integers, and suppose r is the nonzero remainder when b is divided by a . Prove that when $-b$ is divided by a , the remainder is $a - r$.

Solution.

We are given:

- a, b are positive integers.
- r is the **nonzero remainder** when b is divided by a .

By the division algorithm,

$$b = qa + r$$

for some nonnegative integer q , where

$$0 < r < a.$$

We want to find the remainder when $-b$ is divided by a .

Step 1: Negate the equation

Starting with

$$b = qa + r,$$

multiply both sides by -1 :

$$-b = -qa - r.$$

Rewrite this as

$$-b = -(q + 1)a + (a - r).$$

Step 2: Check that $a - r$ is a valid remainder

Since

$$0 < r < a,$$

subtracting from a gives

$$0 < a - r < a.$$

Therefore, $a - r$ satisfies the definition of a remainder: it is nonnegative and less than a .

Thus,

the remainder when $-b$ is divided by a is $a - r$.

Example

Suppose

$$b = 17, \quad a = 5.$$

Then

$$17 = 3(5) + 2,$$

so $r = 2$.

For -17 ,

$$-17 = -4(5) + 3.$$

The remainder is 3, and indeed

$$a - r = 5 - 2 = 3.$$

Hence the theorem holds.